

Global Valuation

Cross-Country Price-Earnings Multiples

Augustin Landier

February 2026

HEC Paris

Outline

Data and Sample

Country Dominates P/E

What Explains Cross-Country P/E?

A Profitability Puzzle

Regional Patterns

Summary and Next Steps

Data and Sample

Data Sources

File	Size	Rows	Cols	Description
compustat_global	2.3 GB	981K	468	Global annual fundamentals (122 countries)
compustat_america	0.8 GB	254K	984	North America fundamentals
returns_global	1.9 GB	10.4M	25	Global monthly security prices

- **Combined panel:** 743,087 firm-years, 72,720 firms, 81 countries (1986–2024)
- All financial data in *local currency* (millions)
- **Market cap (Global):** $cshoi \times prccm$ using $prirow$ (primary security)

Variable Definitions

Dependent variable:

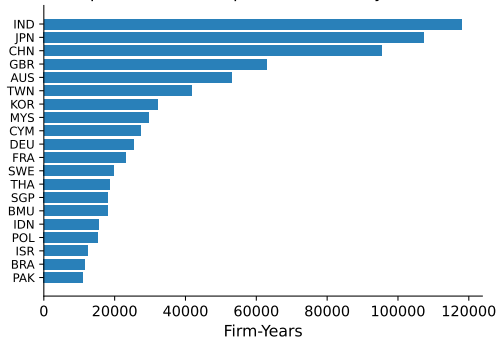
$$\log(P/E)_{i,t} = \log\left(\frac{\text{Market Cap}_{i,t}}{\text{Net Income}_{i,t}}\right), \quad NI > 0$$

Market Cap	US: mkvalt. Global: cshoi $\times$ prccm (primary security)
Net Income	ib (income before extraordinary items)
ROA	ib / at
Leverage	(at - ceq)/at
Industry	GICS Industry Group (~25 categories)

P/E restricted to profitable firms (72% of sample). All ratios winsorized at 1st/99th percentiles.

Coverage and Descriptive Statistics

Compustat Global: Top 20 Countries by Firm-Years



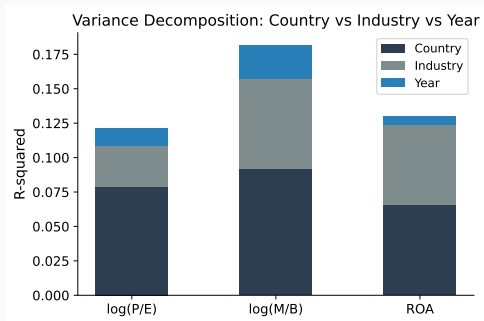
Variable	N	Median	P10	P90
log(P/E)	532K	2.79	1.76	4.32
ROA	743K	0.02	-0.15	0.11
Leverage	743K	0.49	0.14	0.85

Top 5 by obs: USA (188K), JPN (60K), CHN (58K), IND (57K), TWN (31K).

Global prices start 2007.

Country Dominates P/E

Variance Decomposition: Country vs Industry vs Year



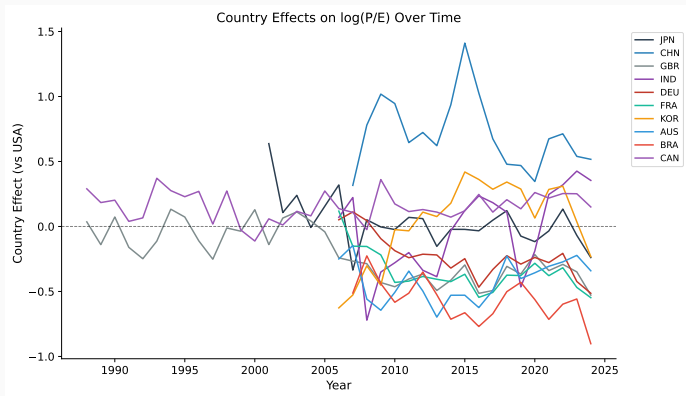
Shapley R^2 decomposition of $\log(P/E)$:

Factor	Share of R^2
Country	64.6%
Industry	24.8%
Year	10.6%

Country alone explains nearly two-thirds of the variation in P/E .

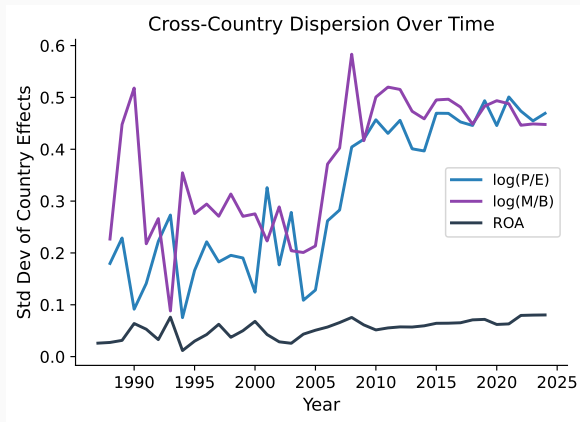
Extends Heston–Rouwenhorst (1994) from *returns* to *valuation levels*.

Country P/E Effects Over Time (vs USA)



Year-by-year regressions with size, leverage, earnings growth, industry FE. USA = 0. China premium largest but declining. Japan near zero. India rising since 2015.

P/E Dispersion Is Diverging



P/E dispersion:

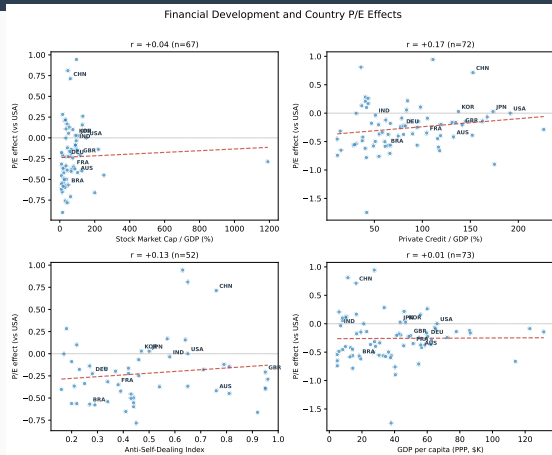
$$\sigma = 0.17 \rightarrow 0.47 \quad (\times 2.7)$$

Despite financial globalization, country valuations are *not converging*.

The gap between the most and least expensive countries has nearly tripled since the early 2000s.

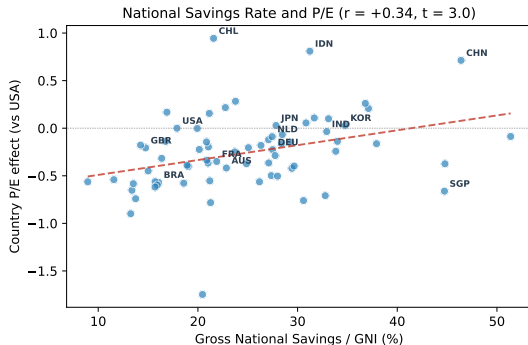
What Explains Cross-Country P/E?

Standard Institutional Variables Do Not Predict P/E



Market cap/GDP ($r = +0.04$), credit/GDP ($r = +0.17$), anti-self-dealing ($r = +0.13$), GDP per capita ($r = +0.01$): all near zero. La Porta et al. (2002) variables do not explain valuation levels.

National Savings Rate Predicts P/E



Gross savings / GNI (World Bank):

$r = +0.34$, $t = 3.01^{***}$ ($n = 70$)

Robust:

- Excl. Gulf states: $t = 3.40$
- Controlling for ROA: $t = 2.85$
- Controlling for GDP/cap: $t = 2.43$

Interpretation: countries that save more have more capital seeking equity returns, pushing up price multiples. A *demand-side* channel.

Orthogonal to profitability ($r = -0.12$).

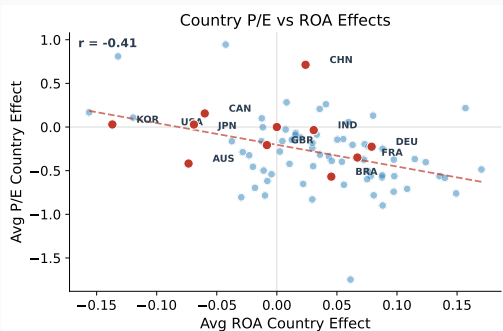
Summary: What Predicts Country P/E?

Variable	Corr with P/E	t-stat	N
Savings / GNI	+0.34***	3.01	70
Private Credit / GDP	+0.17	1.30	72
Anti-Self-Dealing Index	+0.13	0.92	52
Stock Market Cap / GDP	+0.04	0.29	67
GDP per capita (PPP)	+0.01	0.09	73
Pension Fund Assets / GDP	+0.14	1.16	56
Equity Market Depth	~0	—	various

Savings/GNI is the **only** significant predictor of cross-country P/E among standard macro and institutional variables.

A Profitability Puzzle

High-P/E Countries Have Lower Profitability



Country ROA vs P/E: $r = -0.41$

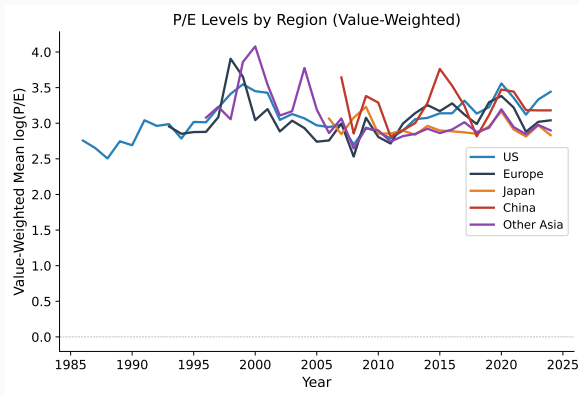
- Not mechanical: lagged ROA gives $r = -0.37$
- Not industry composition: robust with/without industry FE
- Earnings growth uncorrelated ($r \approx 0$)

Within countries, the sign flips ($r = +0.80$).

Consistent with a *discount rate* rather than cash flow story. Details in appendix.

Regional Patterns

Regional P/E Over Time

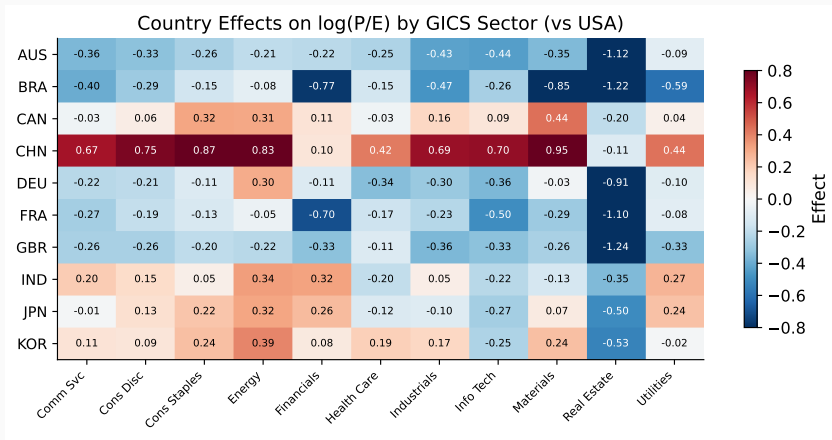


Key patterns:

- US P/E premium is recent and driven by large firms (tech)
- China P/E volatile, declining since 2015
- Japan near Europe

Value-weighted averages. VW gaps larger than EW — US premium concentrated in mega-caps.

Industry × Country P/E Effects



Country P/E effects by GICS sector. The China premium is broad-based across industries. Real estate shows large discounts in most countries.

Summary and Next Steps

Stylized Facts

1. **Country dominates P/E** — 65% of R^2 (Shapley). Extends Heston–Rouwenhorst (1994) from returns to valuation levels.
2. **P/E dispersion is diverging** — $\times 2.7$ since early 2000s, despite financial globalization.
3. **National savings rate predicts P/E** ($r = +0.34$, $t = 3.01$). More savings $\Rightarrow$ more capital chasing equities $\Rightarrow$ higher multiples. The only significant macro predictor.
4. **Standard institutional variables are null** — La Porta, financial development, GDP per capita: all insignificant.
5. **High-P/E countries are less profitable** ($r = -0.41$). Consistent with a discount rate channel rather than a cash flow story.

Research Questions

1. **Why is P/E dispersion diverging?** ($\times 2.7$) — US tech? Post-GFC monetary policy? Capital flows?
2. **Can the savings channel be formalized?** — Savings/GNI predicts P/E and is orthogonal to ROA. A model where savings rates affect discount rates?
3. **Do low-P/E countries earn higher returns?** — Bekaert–Harvey, Fama–French international. If so, valuation gaps reflect risk premia.
4. **What explains China's P/E premium?** — Highest P/E among large countries. Retail investors? Capital controls? Savings rate (46% of GNI)?

Related Literature

Country vs industry effects:

- Heston–Rouwenhorst (1994, *JFE*) — country dominates returns. We extend to *levels*.
- Griffin–Karolyi (1998, *JFE*) — industry effects vary across sectors.

Institutions and investor protection:

- La Porta et al. (2002, *JF*) — investor protection raises Tobin's Q. Our results: weak for P/E.
- Hail–Leuz (2006, *JAR*) — institutions reduce cost of equity by up to 200 bps.
- Stulz (2005, *JF*) — “twin agency problems” prevent equalization.

Savings and asset prices:

- Abel (2003, *Brookings*) — effect of baby boomers' saving on asset prices.
- Caballero et al. (2008, *JME*) — global imbalances and asset price inflation.

Appendix

P/E Regression Specification

Main specification (profitable firms, $NI > 0$):

$$\log(P/E)_{i,t} = \beta_1 \text{Lev} + \beta_2 \log(\text{Assets}) + \beta_3 \Delta E_{t-1} + \gamma_c + \delta_j + \tau_t + \varepsilon_{i,t}$$

	Spec 1	Spec 2
Leverage	−0.017 (0.102)	−0.013 (0.100)
$\log(\text{At})$	−0.076** (0.030)	−0.078** (0.031)
Lag ΔE	−0.016*** (0.003)	−0.016*** (0.003)
FE	Country + Ind + Year	Country × Year + Ind
N / R^2	428K / 0.142	428K / 0.163

SE clustered by country. Year-by-year regressions extract γ_c with $\text{USA} = 0$.

Profitability Puzzle: Robustness Details

Specification	$r(\text{ROA}, \text{P/E})$	Note
Raw country averages	−0.02	No controls
Industry×year demeaned	+0.11	Industry removed
Regression without industry FE	−0.46	Size + leverage + earn. growth Main specification Breaks mechanical link
Regression with industry FE	−0.41	
Using lagged ROA_{t-1}	−0.37	
With USD size controls:		
Both P/E and ROA control for size	−0.64	Strongest
No size controls on either side	−0.02	Disappears

The negative correlation is a **Simpson's paradox**: within each size bucket, high-ROA countries have lower P/E. Without size controls, this is masked. Effect is stronger with USD-converted assets (−0.64 vs −0.51 local), ruling out a currency artifact.

Market Cap Construction (Detail)

US firms: `mkvalt` (pre-computed market value in USD millions). Available from 1986.

Non-US firms (Compustat Global): no pre-computed market value.

- Compute: $\text{Market Cap} = \text{cshoi}$ (shares outstanding) $\times$ `prccm` (month-end price)
- **Pitfall:** firms have multiple securities across exchanges and currencies

Example — SAP SE (`gvkey=103487`) in Dec 2022:

`iid=02W` `EUR` 96.39 ← correct

`iid=05W` `HUF` 38,620 ← wrong currency!

`iid=06W` `MXN` 2,009 ← wrong currency!

- **Solution:** use `prirow` field (primary security designation)

All data in local currency (millions). Cross-country *levels* absorbed by country FE.

Accounting Standards

Compustat Global reports accounting standard via acctstd:

Standard	Share
IFRS	59.7%
Domestic GAAP	39.3%
US GAAP	0.2%
Other / missing	0.8%

US firms: all US GAAP.

Country variation:

- EU: mandatory IFRS since 2005
- Japan: 93% domestic GAAP
- China: 96% domestic GAAP
- Korea: 99% IFRS (since 2011)

P/E is less sensitive to accounting standards than book-equity-based ratios. Country FE absorb time-invariant differences.